目录

Cellometer K2 双荧光细胞分析仪的使用与维护

1 目的

本 SOP 的目的是规范 Cellometer K2 双荧光细胞分析仪的使用、维护及故障排除的方法。

2 适用范围

本 SOP 适用于本机构所有使用 Cellometer K2 双荧光细胞分析仪的人员。

3 职责

职责	职责描述
操作人员	• 应当掌握并遵守本 SOP
仪器负责人	• 负责按照本规程操作维护仪器
管理员	• 负责创建账户并分配用户权限、建立项目、分配项目相关人员

4 定义

无。

5 操作程序

5.1 操作方法

（1）USB 数据线连接电脑，插上电源，打开仪器背后开关，仪器前方指示灯变亮并伴有声音。

（2）双击电脑桌面上的 Cellometer Image Cytometer 软件图标启动程序，弹出软件操作界面。

（3）撕去计数板上下两层保护膜，加入所测样品约 20 μL，将需要计数的一边插入进样口。

（4）依据实际检测类型选择 Assay Type：

假如：如果 Car-T（或者 T 细胞）双荧光 AOPI 计数，点击“SETUP”中“Assay”栏的下拉菜单，选择实验类型“Car-T”;

（5）输入样品名称。

（6）点击“Preview（Preview Brightfield Image）”。

（7）转动仪器右侧下方旋钮调整焦距。

（8）细胞需中心透亮，轮廓清晰。

（9）点击下方计数按钮

进行自动拍摄图片并计数。

（10）显示分析结果。

Results:

Count	Concentration	Mean Diameter
Total Cells: 2412	5.25x10^7 cells/mL	3.5 micron
Live: 1852	4.03x10^7 cells/mL	3.6 microns
Dead: 560	1.21x10^7 cells/mL	3.1 micron

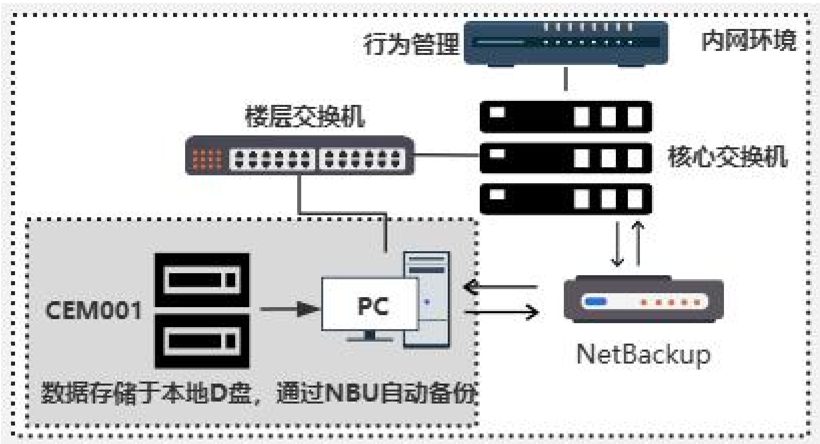
Viability: 76.9%

（11）下一个样品计数：拔出计数板，如有下一个需要计数的样品，重复步骤 3-10。

（12）关机：点击主页面右上角退出程序，关闭仪器背后开关，并填写仪器设备使用维护记录表（SMAN-024-F6）。

5.2 拓扑图

本系统网络拓扑图如下：



5.3 账户管理

该仪器使用本地账号登录，账户类型包括管理员、操作员账户，每个用户由管理员分配唯一的用户名并设定密码，并使用自己的用户账户登录仪器操作，填写仪器使用记录。用户账户的密码需至少每三个月内修改一次且不能重复用最近三次的密码。如果 5 分钟之内，系统没有发生任何操作，系统将自动锁定。

(1) 管理员：管理员通常由设施管理部成员担任，负责创建账户并分配用户权限等。

(2) 操作员：操作员账户仅允许进行编辑、保存结果等操作。

5.4 数据保存和备份

(1) 数据保存路径为：D:\K2 Data。

(2) 对仪器的电子数据进行至少每 10 天 NBU 全量备份。

(3) 本仪器原始记录为纸质记录。

5.5 数据恢复

当数据损坏需要进行数据恢复时，应联系管理员，先进行当前数据库的备份，然后将需要恢复的数据库文件拷贝到指定的保存位置，恢复损坏的文件，并做好记录。

5.6 注意事项

（1）使用前务必确认电源数据线连接完毕，仪器电源接通，再打开软件。

（2）计数板上下两层保护膜都要撕掉，不然会导致视野背景过脏，影响计数结果。

（3）计数前尽量混匀样本，建议反复吹打 10 次左右后取样，以保证计数准确。

（4）PI 染料与样品混合后无需反应时间，即刻可以计数。

（5）吸取样品或染料时注意更换枪头，样品和染料须在细胞板中混合均匀后取样。

5.7 维护保养

（1）每次使用完毕后对细胞计数仪的外表面进行清洁。

（2）若长期不使用时，至少每半年开机一次检查系统的状态，软件、硬件是否能正常运行，并对检查情况进行记录，填写仪器设备使用维护记录表（SMAN-024-F6）。

5.8 性能确认

每年应对仪器进行一次性能确认（PQ），验证资料按要求保存及归档。

5.9 故障处理

若系统发生配件损坏，系统故障等情况，需参照 SOP SMAN-024《仪器设备的管理》和 SMAN-062《计算机化系统应急管理》进行管理和维修，填写 SMAN-024-F8《仪器设备维修记录表》和 SMAN-062-F1《计算机化系统异常汇报、处理记录表》。

6 相关文件

文件编号	SMAN-024	文件标题	仪器设备的管理
	SMAN-062		计算机化系统应急管理
表格编号	SMAN-024-F6	表格标题	仪器设备使用维护记录表
	SMAN-062-F1		计算机化系统异常汇报、处理记录表
	SMAN-024-F8		仪器设备维修记录表

7 参考文献

(1) 21 CFR Part58 Good Laboratory Practice for Nonclinical Laboratory Studies
(2) OECD Good Laboratory Practice for Nonclinical Laboratory Studies
(3) 中国国家食品药品监督管理局“药物非临床研究质量管理规范”

8 SOP修订历史

修订历史

SOP 现版本号	旧版本生效日期	修订者	修订内容描述
V2.0	2024-07-09	李玉薇	1、SOP 双语化 2.“5.2 注意事项”增加（6）细胞计数的电子数据需随试验进行归档。
V3.0	2024-10-17	李玉薇	1.增加 3 管理员的职责、增加“5.2 账号管理、5.3 项目的创建和审计追踪、5.4 数据保存和备份、5.5 数据恢复”内容。
V4.0	2024-11-21	李玉薇	1. 修改仪器名称为 Cellometer K2 双荧光细胞分析仪 2. 5.4 数据保存和备份时间修改为至少每月。 3. 5.4 增加“本仪器原始记录为纸质记录”。
V5.0	2025-02-27	李玉薇	1. 删除“5.3 项目的审计追踪”。
V6.0	2025-04-24	李玉薇	1. 增加“5.2 拓扑图”。 2. 修改“5.4 数据保存和备份”对仪器电子数据的备份。 3. 修改“5.9 故障处理”的处理方式。
V7.0	2025-06-20	李玉薇	1. 删除“5.3 账户管理”（1）管理员建立项目、分配项目相关人员和“5.6 注意事项”（6）细胞计数的电子数据需随试验进行归档。

Use and Maintenance of Cellometer K2 Fluorescent Cell Counter

1 Purpose

The purpose of this SOP is to standardize the methods for using, maintaining, and troubleshooting of the Cellometer K2 Fluorescent Cell Counter.

2 Scope

This SOP is applicable to all personnel using the Cellometer K2 Fluorescent Cell Counter at this facility.

3 Role

Role	Responsibility
Operators	• Must understand and comply with this SOP.
Equipment Manager	• Responsible for operating and maintaining the equipment according to this SOP.
Administrator	• Responsible for creating accounts and assigning user rights, creating projects, assigning project-related personnel.

4 Definitions

NA.

5 Procedures

5.1 Operating Procedure

(1) Connect the USB data cable to the computer, plug in the power, turn on the switch at the back of the equipment. The front indicator light of the equipment will illuminate accompanied by a sound.
(2) Double-click the Cellometer Image Cytometer software icon on the computer desktop to launch the program and open the software interface.
(3) Peel off the protective films from both sides of the counting chamber, add about 20 μL of the sample to be measured, and insert the side needing counting into the sample port.
(4) Select the Assay Type based on the actual assay type:
For example: For Car-T (or T cells) dual fluorescence AO/PI counting, click on "Assay" in the "SETUP" tab, choose "Car-T" from the drop-down menu.

(5) Enter the sample name. 

(6) Click "Preview Brightfield Image". 

(7) Rotate the knob at the bottom right of the equipment to adjust focus.

(8) Cells should appear centered and bright, with clear outlines.

(9) Click the counting button 

below to automatically capture images and count.

(10) Display the analysis results.

Results:

Count	Concentration	Mean Diameter
==============	==============	==============
Total Cells: 2412	5.25x10^7 cells/mL	3.5 micron
Live: 1852	4.03x10^7 cells/mL	3.6 microns
Dead: 560	1.21x10^7 cells/mL	3.1 micron

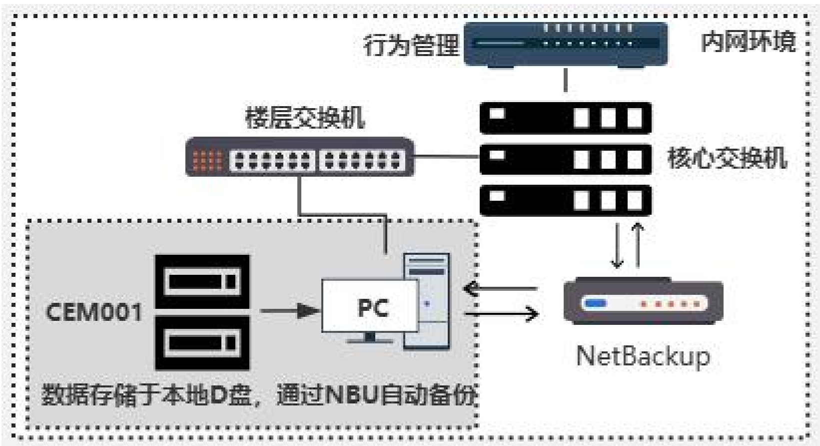
Viability: 76.9%

(11) Counting the next sample: Remove the counting chamber. If another sample needs to be counted, repeat Steps 3-10.

(12) Shutdown: Click the  in the upper right corner of the main page to exit the program, turn off the switch at the back of the equipment, and complete the Equipment Use and Maintenance Record (SMAN-024-F6).

5.2 Topological map

The network topology of this system is as follows:



5.3 Account management

The equipment uses a local account to log in, and account types include administrator and operator accounts. Each user is assigned a unique user name and set a password by the administrator, and uses his or her own user account to log in to the instrument to operate and fill in the equipment 's usage records. The password for the user account must be changed once at least every three months and the last three passwords cannot be reused. If no operation occurs within 5 minutes, the system will automatically lock.

(1) Administrator: The administrator account is usually held by a member of facilities operation department and is responsible for creating accounts and assigning user rights, etc.

(2) Operator: The operator account only allows operations such as editing and saving results.

5.4 Data retention and backup

(1) The data save path is: D:\K2 Data.

(2) Perform full NBU backups of the equipment's electronic data at least every 10 days.

(3) The original records of this equipment are paper records.

5.5 Data recovery

When data corruption requires data recovery, you should contact the administrator to make a backup of the current database, then copy the database files to be recovered to the designated save location, recover the corrupted files, and make a record.

5.6 Notes

(1) Before using, ensure the USB data cable is properly connected, the equipment is powered on, and open the software again.

(2) Remove both protective films from the counting chamber to prevent a dirty background affecting counting results.

(3) Before counting, thoroughly mix the sample; it is recommended to pipette and mix approximately 10 times to ensure accurate counting.

(4) PI dye can be counted immediately without reaction time after mixing with the sample.

(5) When aspirating samples or dyes, ensure to change the pipette tips; samples and dyes should be thoroughly mixed in the cell chamber.

5.7 Maintenance

(1) Clean the external surface of the cell counter after each use.

(2) If not used for an extended period, power on the equipment at least every six months to check system status, software, and hardware functionality. Record and complete the Equipment Use and Maintenance Record (SMAN-024-F6) accordingly.

5.8 Performance Qualification

Perform the performance qualification (PQ) annually to validate equipment performance. Ensure validation data is stored and archived as required.

5.9 Troubleshooting

In case of damage to accessories or system failure, the system should be managed and repaired with reference to SOP SMAN-024 Equipment Management and SMAN-062 Computerized System Emergency Management, and fill in SMAN-024-F8 Equipment Repair Record and SMAN-062-F1 Computerized System Exception Reporting and Processing Record.

6Relevant Documents

Document Number	SMAN-024	Document Title	Management of Equipment
Form Number	SMAN-024-F6	Form Title	Equipment Use and Maintenance Record
	SMAN-062-F1		Computerized System Exception Reporting and Processing Record
	SMAN-024-F8		Equipment Repair Record
Template Number	NA	Template Title	NA

7 References

(1) The United States Food and Drug Administration (US FDA) Good Laboratory Practice for Nonclinical Laboratory Studies (21 CFR Part 58).
(2) OECD Good Laboratory Practice for Nonclinical Laboratory Studies.
(3) National Medical Products Administration (NMPA) Good Laboratory Practice for Nonclinical Laboratory Studies.

8 SOP Revision History

Revision History

SOP current version No.	Effective date of the old version	Reviser	Description of revision
V2.0	2024-07-09	Yuwei Li	1. Bilingual SOP 2. “5.2 Notes” Add (6) Electronic data from cell counts need to be archived with the test.
V3.0	2024-10-17	Yuwei Li	1.Add 3 administrator of role, Add 5.2 Account management, 5.3 Project creation and audit trail, 5.4 Data retention and backup, 5.5 Data recovery
V4.0	2024-11-21	Yuwei Li	1. Changed the name of the equipment to Cellometer K2 Fluorescent Cell Counter. 2. 5.4 Data retention and backup time

			modified to at least once a month. 3. 5.4 Add "the original records of this equipment are paper records".
V5.0	2025-02-27	Yuwei Li	1. Delete "5.3 Project audit trail"
V6.0	2025-04-24	Yuwei Li	1. Add "5.2 Topological map". 2. Modify the backup of electronic data of the equipment in "5.4 Data retention and backup". 3. modify the handling method of "5.9 Troubleshooting".
V7.0	2025-06-20	Yuwei Li	1. Delete "5.3 Account management "(1) Administrator: creating projects, assigning project-related personnel and "5.6 Notes" (6) Electronic data from cell counts need to be archived with the test.